UNRAVELLING THE COMPLEXITY OF COMORBIDITIES AND POLYPHARMACY IN EPILEPTIC PATIENTS FROM PATIENT-GENERATED DATA: A MACHINE LEARNING DRIVEN STUDY

By

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Abstract:

Objectives:

The primary objective of this research is to unravel the complexities of comorbidities and polypharmacy in epileptic patients using data from NHANES 2013-2014 and 2015-2016 cycles. Specifically, I aim to: Identify epilepsy patients through systematic screening of antiepileptic drug prescriptions and medical condition questionnaires, capturing 828 patients across both identification methods; Analyze the prevalence and patterns of 13 major comorbidities among epilepsy patients, including cardiovascular, metabolic, and psychiatric conditions; Quantify polypharmacy burden by categorizing patients into low (1-4 medications), moderate (5-9 medications), and high (10+ medications) risk groups; Compare comorbidity and medication patterns between epilepsy patients and the general population using statistical tests including Mann-Whitney U and Fisher's exact tests; Calculate odds ratios for individual comorbidities to identify conditions significantly associated with epilepsy; Develop risk stratification models and conduct machine learning clustering analysis to identify distinct patient phenotypes within the epilepsy population; Prepare a comprehensive dataset for machine learning analysis with 17 key features to enable predictive modeling. The ultimate goal is to provide data-driven insights that can inform clinical decision-making and improve care quality for epilepsy patients managing multiple chronic conditions and complex medication regimens.

Methods: This retrospective cross-sectional study utilized data from the National Health and Nutrition Examination Survey (NHANES) 2013-2014 and 2015-2016 cycles. I loaded six core NHANES datasets using pandas read_sas function: demographics (DEMO_H: n=10,175; DEMO_I: n=9,971), medical conditions questionnaire (MCQ_H: n=9,770; MCQ_I: n=9,575), and prescription medications (RXQ_RX_H: n=20,194; RXQ_RX_I: n=19,647). Data integration was performed by merging datasets using respondent sequence numbers (SEQN), resulting in a combined dataset of 20,146 unique participants.Epilepsy patients were identified through two complementary approaches: (1) antiepileptic drug (AED) usage by searching the RXDDRUG field for a comprehensive list of 46 AED medications including both generic and brand names, and (2) medical conditions questionnaire responses (MCQ160E) indicating physician-diagnosed epilepsy. The AED search identified 497 prescriptions in 2013-2014 and 492 in 2015-2016, while the medical questionnaire approach identified 481 patients, representing a final cohort of 828 unique epilepsy patients. Comorbidity analysis examined 13 conditions using MCQ variables: MCQ160A (Arthritis), MCQ160B (Congestive Heart Failure), MCQ160C (Coronary Heart Disease), MCQ160D (Angina), MCQ160E (Heart Attack), MCQ160F (Stroke), MCQ160G (Emphysema), MCQ160K (Chronic Bronchitis), MCQ160L (Liver Condition), MCQ160M (Thyroid Problem), MCQ160N (Kidney Disease), MCQ220 (Cancer), and MCQ240A (Diabetes). Polypharmacy was assessed by counting total medications per patient and categorizing into: no medications, low (1-4), moderate (5-9), and high (10+). Statistical analyses included Mann-Whitney U tests for continuous variables, Fisher's exact tests for categorical comparisons, and Spearman correlation analysis for comorbidity-medication relationships. Machine learning techniques included K-means clustering with PCA dimensionality reduction to identify distinct patient phenotypes within the epilepsy cohort. A composite risk stratification score was developed incorporating age (≥65 years: +2 points), comorbidity burden (≥4 conditions: +3 points), and polypharmacy status (≥10 medications: +3 points).

Results:

A total of 20,146 participants were included across the two NHANES cycles, of whom 828 were identified as epilepsy patients. Within the epilepsy cohort specifically (n=481 from medical questionnaire responses), patients demonstrated exceptionally high comorbidity burden with 100% reporting at least one additional chronic condition

beyond epilepsy. The mean comorbidity count was 3.63 ± 1.74, with heart attack affecting 100% of patients, arthritis in 55.9%, and coronary heart disease in 47.2%. A strong positive correlation ($r=0.476$, $p<0.001$) was observed between comorbidity count and total medications, indicating that patients with more chronic conditions require proportionally more medications. Age stratification revealed that elderly patients (≥65 years, n=297, 61.7%) had higher medication burden (6.78 vs 5.71 medications) and slightly more comorbidities (3.72 vs 3.47) compared to younger adults. Machine learning clustering analysis identified three distinct epilepsy patient phenotypes: Cluster 0 (n=87) representing older patients with moderate comorbidity burden (age 74.4, 4.2 comorbidities), Cluster 1 (n=99) showing high-complexity patients (age 65.4, 5.9 comorbidities), and Cluster 2 (n=295) comprising the majority with lower complexity (age 65.3, 2.7 comorbidities). Risk stratification analysis classified 75.5% of epilepsy patients as high-risk, 19.5% as moderate-risk, and only 5.0% as low-risk based on the composite scoring system. Gender analysis revealed that women had slightly higher comorbidity burden (3.95 vs 3.44) and medication counts (6.63 vs 6.22) compared to men, though both groups showed similar age distributions. Comparison with the general population demonstrated that epilepsy patients had significantly higher rates across all measured comorbidities, with odds ratios ranging from 3.7 (coronary heart disease) to 8.5 (arthritis). The prepared machine learning dataset included 17 variables encompassing demographics, comorbidities, and medication burden, providing a robust foundation for future predictive modeling applications.

Conclusion:

This comprehensive study demonstrates that epilepsy patients in the United States experience extraordinarily high rates of multimorbidity and polypharmacy compared with the general population, with distinct patterns that vary by age, gender, and risk profile. The findings reveal a complex healthcare landscape where seizure management intersects with treatment for cardiovascular, metabolic, and respiratory comorbidities, creating substantial polypharmacy risk that affects nearly two-thirds of patients. The identification of three distinct patient clusters through machine learning analysis suggests opportunities for personalized care approaches tailored to specific phenotypes. The composite risk stratification tool successfully classified patients into

meaningful categories that could guide clinical resource allocation and intervention strategies.

Most significantly, the universal presence of comorbidities in the epilepsy cohort (100% prevalence) and the strong correlation between comorbidity burden and medication complexity (r=0.476) underscore the critical need for integrated, multidisciplinary care models. These results provide essential evidence for healthcare policy development, clinical practice guidelines, and precision medicine initiatives aimed at optimizing treatment outcomes while minimizing adverse effects from polypharmacy. The comprehensive datasets and analytical frameworks developed in this study offer valuable resources for future research in epilepsy care optimization and population health management.

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1.Introduction:

Epilepsy stands as one of the most prevalent neurological disorders globally, affecting approximately 50 million individuals worldwide and representing a significant public health challenge that transcends geographical, economic, and demographic boundaries (World Health Organization, 2024). The World Health Organization estimates that nearly 80% of those affected reside in low- and middle-income countries, where three-quarters do not receive the necessary treatment, highlighting profound disparities in access to care. Despite the fact that up to 70% of individuals with epilepsy could achieve seizure freedom with appropriate diagnosis and treatment, the risk of premature death remains up to three times higher than in the general population. Beyond the clinical challenges, stigma and discrimination continue to impact individuals with epilepsy and their families across diverse cultural contexts, creating additional barriers to optimal care and quality of life.

The complexity of epilepsy management extends far beyond seizure control, encompassing a intricate web of comorbid conditions that frequently accompany this neurological disorder. Comorbidities, defined as medical conditions that coexist with a primary diagnosis, fundamentally alter treatment approaches and clinical outcomes. Recent epidemiological data from the United States reveals that over 84% of hospital patients in 2019 presented with at least one comorbidity, with 50% having three or more concurrent conditions. The most prevalent comorbidities identified include hypertension (53.2%), diabetes (26.4%), and chronic lung disease (20.5%). The impact of age on comorbidity burden is particularly striking, with only 8% of individuals aged 18-24 years having three or more comorbidities, compared to 73% of those aged 85 and older. The clinical implications are severe, with surgery-related mortality being eight times higher for patients with three or more comorbidities compared to those without significant comorbid conditions.

Polypharmacy, conventionally defined as the concurrent use of five or more medications, represents another critical dimension of complexity in modern healthcare. This phenomenon is increasingly common among older adults and at-risk populations due to the rising prevalence of multiple chronic conditions requiring pharmaceutical intervention. While polypharmacy may be clinically necessary and appropriate in many cases, excessive medication use can lead to serious adverse outcomes including increased risk of falls, frailty, cognitive impairment, and mortality.

The historical perspective on polypharmacy has evolved significantly, transitioning from a focus primarily on unnecessary drug use to modern approaches that emphasize evidence-based prescribing and the reduction of inappropriate medication use. Contemporary challenges in polypharmacy management include adverse drug effects, complex drug-drug interactions, medication non-adherence, and prescribing cascades where new medications are prescribed to treat side effects of existing drugs.

The intersection of epilepsy, comorbidities, and polypharmacy creates a particularly complex clinical scenario that demands sophisticated analytical approaches. Machine learning has emerged as a powerful tool for understanding and predicting drug-drug interactions (DDIs), representing a key challenge in managing patients with multiple chronic conditions requiring polypharmacy. Recent computational advances have demonstrated the potential of integrating molecular, genomic, and health record data to predict DDIs with high accuracy. By utilizing large-scale datasets that include drug-gene interactions and patient medical records, these approaches offer promising solutions for improving the management of comorbidities and polypharmacy, particularly in situations where multiple medications are prescribed simultaneously.

The National Health and Nutrition Examination Survey (NHANES) provides an exceptional opportunity to investigate these complex relationships using nationally representative data from the United States population. NHANES represents a long-standing program initiated by the Centers for Disease Control and Prevention (CDC) and the National Center for Health Statistics (NCHS) that systematically collects health-related data from the noninstitutionalized civilian population. The survey combines structured household interviews with standardized physical examinations conducted in mobile health centers, creating a uniquely comprehensive source of information that encompasses demographics, medical histories, laboratory measurements, and detailed medication records.

In this comprehensive study, I employ a sophisticated analytical approach using NHANES data from 2013-2014 and 2015-2016 cycles to investigate comorbidities and polypharmacy patterns in epileptic patients. My methodology encompasses multiple complementary strategies: loading and combining six key NHANES datasets including demographics (DEMO_H and DEMO_I), medical conditions questionnaire (MCQ_H and MCQ_I), and prescription medications (RXQ_RX_H and RXQ_RX_I). I

implement dual epilepsy patient identification strategies using both antiepileptic drug (AED) usage patterns and medical condition questionnaire responses, identifying patients taking medications including gabapentin, clonazepam, lorazepam, topiramate, lamotrigine, levetiracetam, phenytoin, carbamazepine, and other commonly prescribed AEDs. Through this comprehensive approach, I identified 828 unique epilepsy patients across both survey cycles for comparative analysis.

The analytical framework includes comprehensive comorbidity analysis examining 13 major chronic conditions including arthritis, congestive heart failure, coronary heart disease, stroke, chronic kidney disease, and various other systemic conditions. My polypharmacy assessment categorizes patients into clinically meaningful risk groups based on medication counts, with particular emphasis on those taking five or more medications concurrently. I develop sophisticated statistical comparisons using Mann-Whitney U tests and Fisher's exact tests to evaluate differences between epilepsy and non-epilepsy populations, while also conducting focused analyses within the epilepsy cohort itself. Advanced machine learning techniques including K-means clustering with principal component analysis enable identification of distinct patient phenotypes within the epilepsy population. Finally, I prepare comprehensive datasets for future machine learning analysis, creating feature matrices with 17 variables encompassing demographics, comorbidities, and medication patterns to enable advanced predictive modeling applications.

2.Literature review:

2.1 Summary Review of Literature:

Recent studies have explored multiple aspects of epilepsy, including its comorbidities, treatment challenges, medication adherence, and economic burden. Machine learning (ML) has emerged as a promising tool in precision medicine for epilepsy management. (Linden et al., 2021) addressed challenges related to limited data availability and model interpretability by developing DeepLORI, a multimodal neural network that mapped ICD codes to PheWAS codes for enhanced granularity. This model significantly outperformed traditional approaches in predicting six common epilepsy-related comorbidities and demonstrated reliable predictions up to six years in advance.

The integration of SHAP (Shapley Additive Explanations) provided individual-level interpretability, a crucial aspect in clinical settings to foster trust among healthcare

providers and patients. By identifying high-risk individuals early, such models can contribute to preventive interventions and personalized therapy adjustments. Despite advancements in deep learning models for epilepsy detection and seizure prediction using EEG signals, numerous challenges remain. (Zhang et al., 2024) identified key limitations, including a lack of diverse datasets, inadequate preprocessing techniques, inconsistencies in false positive rate evaluations, and difficulties distinguishing between interictal and preictal periods. Current models also suffer from high computational complexity, making real- time implementation challenging. Future research needs to focus on expanding datasets with diverse patient demographics, improving model interpretability, refining seizure prediction parameters, and optimizing algorithms for real-time use in clinical and home-monitoring applications. Multimorbidity and polypharmacy patterns in elderly epilepsy patients were examined in a study conducted in Catalonia (Stafford et al., 2021). The analysis identified six out of seven multimorbidity clusters characterized by an overrepresentation of chronic diseases and associated drug categories. Notably, specific groupings were observed, such as clusters involving diabetes, neurological and musculoskeletal conditions, and cardio-cerebrovascular and renal issues. Moreover, significant sex-based differences were reported, with neurological and musculoskeletal disorders being more prevalent in one sex over the other. These findings provide critical insights for evaluating multimorbid and polymedicated individuals and can inform future clinical guidelines aimed at optimizing treatment strategies for elderly epilepsy patients. Persistence with anti-seizure medications (ASMs) plays a crucial role in epilepsy management, yet adherence remains a major challenge. (Marshall et al., 2022) analyzed data from 6,449 epilepsy patients within the NHS Greater Glasgow and Clyde health board, revealing wide variability in ASM persistence rates, ranging from 11.8% to 78.6%. Younger patients, individuals with a history of non-persistence, and those with fewer comorbidities exhibited lower adherence rates. Conversely, patients with cardiac comorbidities, a history of stroke, or those prescribed newer-generation ASMs demonstrated higher persistence. These findings suggest that targeted interventions, such as patient education programs and adherence monitoring, may be beneficial for younger patients, individuals from socioeconomically deprived backgrounds, and those with a history of non-persistence. The treatment of status epilepticus (SE) with benzodiazepines (BZDs) was retrospectively analyzed in a cohort of 328 patients in Cologne over two years (Kohle et al., 2023). The study found that only 72% of patients with SE received BZDs as

an initial treatment, and merely 21.6% received an adequate dose according to clinical guidelines. Delayed or insufficient BZD administration was more common among older patients, those with less-known epilepsy history, and individuals who arrived later at the emergency department. Adequate BZD dosing was associated with significantly reduced ventilation duration and shorter hospital stays in patients with generalized convulsive SE. However, in cases of non-generalized convulsive SE, aggressive BZD treatment led to longer hospital stays and worsened clinical outcomes, highlighting the necessity for precise dosing strategies and better adherence to SE treatment protocols. Epilepsy also imposes a significant economic burden on healthcare systems, as evidenced by a study in Denmark that utilized national health registers to estimate the prevalence and associated costs of epilepsy (Christensen et al., 2022). As of December 31, 2016, the prevalence of epilepsy was reported at 0.67%, with a slightly higher prevalence in males (0.69%) compared to females (0.65%). Psychiatric comorbidities were notably high, affecting 37.4% of epilepsy patients compared to 15.9% in the general population. The study further estimated the annual individual net cost of epilepsy at €30,683, reflecting direct medical expenses and indirect costs related to lost productivity. Notably, epilepsy prevalence showed an increasing trend among older adults compared to 2006, while pediatric epilepsy cases declined. These findings underscore the pressing need for cost-effective healthcare policies, targeted interventions for psychiatric comorbidities, and improved resource allocation for epilepsy care. Collectively, these studies highlight the growing role of ML in epilepsy prediction, the urgent need for improved medication management in elderly patients, the ongoing challenges in EEG-based seizure prediction, and the impact of multimorbidity and treatment adherence on epilepsy outcomes. The findings also emphasize the necessity for stricter adherence to treatment protocols for SE and the substantial economic burden associated with epilepsy. Addressing these issues through precision medicine, enhanced clinical guidelines, real-time seizure monitoring advancements, and healthcare policy reforms could significantly improve epilepsy management and patient outcomes. The burden of polypharmacy in epilepsy has emerged as a significant clinical concern, with recent studies demonstrating that patients with epilepsy experience substantially higher rates of medication complexity compared to the general population. Gaitatzis and Majeed (2023) established that multimorbidity is nearly universal among epilepsy patients, with cardiovascular, psychiatric, and metabolic conditions driving increased medication requirements. Nicholas et al. (2016) found that 63% of epilepsy patients used five or

more concurrent medications, while Patsalos et al. (2018) identified clinically significant drug interactions in 40% of cases. Central nervous system polypharmacy presents particular risks, as Faught et al. (2009) demonstrated increased rates of cognitive impairment and falls among epilepsy patients taking multiple CNS-active medications. Of special concern are combinations involving opioids and benzodiazepines, which Zaccara et al. (2016) found in 15% of epilepsy patients despite FDA black box warnings regarding respiratory depression. The quality of life implications are substantial, with Biton (2007) showing that 25% of variance in patient-reported outcomes was attributable to medication adverse effects alone. Recent advances in machine learning for drug interaction prediction offer promising solutions, as Vo et al. (2022) demonstrated the potential for computational approaches to identify novel drug-drug interactions affecting epilepsy patients. However, significant gaps remain in understanding optimal medication management strategies for this vulnerable population, particularly regarding evidence-based approaches to reducing inappropriate polypharmacy while maintaining effective treatment of both seizures and comorbid conditions (Samuel W. Terman et al., 2020).

3. Methods:

3.1 Study Design and Dataset:

This investigation employed a comprehensive retrospective, cross-sectional design utilizing secondary analysis of the National Health and Nutrition Examination Survey (NHANES) data. NHANES represents a cornerstone program of health surveillance in the United States, initiated and maintained by the Centers for Disease Control and Prevention (CDC) and the National Center for Health Statistics (NCHS). The program systematically collects nationally representative health-related data from the noninstitutionalized civilian population through a sophisticated combination of structured household interviews and standardized physical examinations conducted in specially equipped mobile health centers.

The NHANES survey design incorporates a multistage probability sampling methodology that ensures robust representativeness of the U.S. population, including strategic oversampling of minority groups and older adults to enable meaningful subgroup analyses. This design feature is particularly valuable for epilepsy research, as it provides adequate sample sizes for examining health

disparities and age-related patterns that might be obscured in smaller or less representative datasets.

For this comprehensive investigation, I utilized two consecutive NHANES cycles: 2013-2014 and 2015-2016. These specific cycles were selected based on several critical factors including temporal proximity, comprehensive availability of prescription medication data, and robust medical condition documentation necessary for accurate identification of individuals with epilepsy and associated comorbidities. The decision to combine two cycles was strategically motivated by the need to increase statistical power and enable more stable estimates of prevalence and associations, particularly important given the relatively low prevalence of epilepsy in the general population. Each NHANES cycle includes approximately 10,000 participants, resulting in a combined analytical dataset comprising over 20,000 individuals. The harmonization process involved careful merging of datasets using the respondent sequence number (SEQN), a unique identifier assigned to each participant that ensures accurate linkage of demographic, health, and medication records across all data collection components.

Three primary NHANES datasets formed the foundation of this analysis: the demographic file (DEMO), containing comprehensive information on age, sex, race, ethnicity, and socioeconomic indicators; the medical conditions questionnaire file (MCQ), documenting whether participants reported physician-diagnosed chronic conditions; and the prescription medication file (RXQ_RX), providing detailed listings of all prescription medications taken by participants in the 30 days preceding the survey, including up to three self-reported indications for each medication.

The integration of these datasets enabled construction of a comprehensive analytical file that facilitated identification of epilepsy cases through multiple methodologies, quantification of comorbidity burden across numerous chronic conditions, and measurement of overall medication load. The NHANES design ensures exceptionally high data quality through rigorous administrator training protocols and verification procedures, including inspection of prescription containers whenever possible, thereby minimizing reporting bias particularly regarding medication use.

3.2 Patient Selection and Epilepsy Identification:

The development of a robust epilepsy identification strategy represented a critical methodological component of this investigation, given the absence of a direct epilepsy diagnosis variable in NHANES. I implemented a dual approach to maximize

case identification while maintaining specificity. The primary method involved identification through antiepileptic drug (AED) usage patterns, while a secondary approach utilized medical condition questionnaire responses.

For the medication-based identification approach, I conducted systematic screening of the RXQ_RX prescription medication files for all antiepileptic drugs. This process involved searching the RXDDRUG field using a comprehensive list of 46 AED medications including both generic and brand names. The AED list encompassed: phenytoin, dilantin, carbamazepine, tegretol, valproic acid, depakote, lamotrigine, lamictal, levetiracetam, keppra, topiramate, topamax, oxcarbazepine, trileptal, gabapentin, neurontin, pregabalin, lyrica, zonisamide, zonegran, lacosamide, vimpat, eslicarbazepine, aptiom, perampanel, fycompa, clobazam, onfi, felbamate, felbatol, rufinamide, banzel, vigabatrin, sabril, tiagabine, gabitril, phenobarbital, luminal, primidone, mysoline, ethosuximide, zarontin, clonazepam, klonopin, diazepam, valium, lorazepam, and ativan.

The medication-based search identified 497 AED prescriptions in the 2013-2014 cycle and 492 prescriptions in the 2015-2016 cycle, representing a substantial pool of potential epilepsy cases. However, recognizing that some of these medications may be prescribed for conditions other than epilepsy (such as neuropathic pain, psychiatric disorders, or migraine prevention), I implemented additional specificity criteria when possible.

The secondary identification approach utilized responses to the medical conditions questionnaire, specifically the MCQ160E variable, which captures whether participants have ever been told by a healthcare professional that they have epilepsy. This direct questioning approach identified 481 individuals across both survey cycles who reported physician-diagnosed epilepsy.

The combination of both identification methods yielded a final epilepsy cohort of 828 unique individuals across the two survey cycles, corresponding to a weighted national prevalence of approximately 1.2% of the U.S. population. This prevalence estimate aligns closely with established epidemiological data for epilepsy, providing validation for the case identification methodology employed.

For analytical purposes, participants not meeting epilepsy criteria through either identification method served as the comparison group, enabling robust statistical comparisons between epilepsy and non-epilepsy populations. The decision to maintain participants with incomplete medication records or missing indication data

required careful consideration, ultimately leading to their exclusion to ensure validity of case classification and maintain analytical integrity.

3.3 Variable Definitions and Data Processing:

The analytical framework required careful definition and processing of numerous variables to support comprehensive investigation of comorbidities and polypharmacy patterns. Demographic variables extracted from the DEMO dataset included age (RIDAGEYR), sex (RIAGENDR), and self-identified race and ethnicity (RIDRETH3), along with socioeconomic indicators such as education level (DMDEDUC2) and family income-to-poverty ratio (INDFMPIR). These variables were considered essential potential confounders of both epilepsy prevalence and comorbidity burden, requiring inclusion in multivariable analyses.

Clinical variables were systematically extracted from the MCQ dataset, focusing on 13 chronic diseases with established clinical relevance and documented associations with epilepsy in the medical literature. The selected conditions included: arthritis (MCQ160A), congestive heart failure (MCQ160B), coronary heart disease (MCQ160C), angina (MCQ160D), heart attack (MCQ160E), stroke (MCQ160F), emphysema (MCQ160G), chronic bronchitis (MCQ160K), liver condition (MCQ160L), thyroid problems (MCQ160M), kidney disease (MCQ160N), cancer (MCQ220), and diabetes (MCQ240A). These conditions were chosen based on their documented associations with epilepsy, their contributions to multimorbidity in aging populations, and their availability with consistent coding across both NHANES cycles.

For each participant, I calculated a composite comorbidity count representing the total number of chronic conditions present, enabling quantitative assessment of multimorbidity burden. Comorbidity burden was further categorized into clinically meaningful groups: no comorbidities, low burden (1-2 conditions), moderate burden (3-4 conditions), and high burden (5 or more conditions). These thresholds were selected based on clinical literature regarding multimorbidity definitions and their associations with healthcare utilization and adverse outcomes.

Medication-related variables represented central components of the analytical framework. For each participant, the RXQ_RX dataset provided comprehensive listings of prescription drugs taken in the preceding 30 days. I derived a total medication count variable serving as the primary measure of polypharmacy burden. Consistent with established clinical literature, polypharmacy was defined as concurrent use of five or more medications, while high polypharmacy was defined as

ten or more medications. These thresholds reflect clinically meaningful levels of medication burden associated with increased risks of drug-drug interactions, adverse events, and poor adherence.

Additional derived variables included binary indicators for polypharmacy and high polypharmacy status, as well as medication counts stratified by therapeutic class when feasible. For epilepsy patients specifically, I identified concurrent use of central nervous system-acting medications including antidepressants, benzodiazepines, and opioids, enabling exploration of CNS polypharmacy patterns that may carry particular clinical significance given potential for overlapping side effects and drug interactions.

3.4 Statistical Analysis Framework:

The statistical approach was designed to comprehensively describe comorbidity and polypharmacy patterns in epilepsy patients relative to the general population, identify significant associations, and explore relationships within the epilepsy cohort itself. All analyses were conducted using Python's scientific computing libraries, including pandas for data manipulation, scipy for statistical testing, and scikit-learn for machine learning applications.

For categorical variables, including presence or absence of specific comorbidities, Fisher's exact test was employed due to its appropriateness for contingency tables with small expected cell counts that can occur with less common conditions. This nonparametric test provides exact p-values regardless of sample size or cell count distributions, making it particularly suitable for the analysis of rare conditions. Statistical significance was defined as p-values less than 0.05.

For continuous variables, particularly total medication counts and comorbidity counts, the Mann-Whitney U test was utilized to compare distributions between epilepsy and non-epilepsy groups. This nonparametric test was selected due to the typically non-normal, right-skewed distributions characteristic of medication and comorbidity count data. Results were reported as medians with interquartile ranges for primary interpretation, supplemented by means and standard deviations for additional context.

Odds ratios with 95% confidence intervals were calculated to quantify the strength of associations between epilepsy status and individual comorbidities. These measures provide clinically interpretable assessments of relative risk and enable comparison of association magnitudes across different conditions. Forest plots were generated to

provide visual summaries of these associations, facilitating rapid interpretation of results.

Within the epilepsy cohort specifically, I conducted correlation analyses using Spearman's rank correlation coefficient to assess relationships between comorbidity burden and medication counts. This nonparametric approach was chosen due to the ordinal nature of count data and the presence of potential outliers in medication counts.

3.5 Advanced Analytics and Machine Learning:

To identify distinct phenotypic patterns within the epilepsy population, I implemented unsupervised machine learning techniques including K-means clustering with principal component analysis (PCA) for dimensionality reduction. The clustering analysis was conducted on standardized features including age, comorbidity count, and individual comorbidity indicators to identify natural groupings of epilepsy patients with similar clinical characteristics.

Prior to clustering, all continuous variables were standardized using z-score normalization to ensure equal weighting across features with different scales. PCA was applied to reduce dimensionality while preserving maximal variance in the data, with the number of components selected based on cumulative explained variance thresholds.

The optimal number of clusters was determined using the elbow method, examining within-cluster sum of squares across different values of k. Once optimal cluster numbers were identified, final clustering was performed and clusters were characterized by examining mean values of key clinical variables within each group.

3.6 Risk Stratification Development

I developed a composite risk stratification framework specifically for the epilepsy population, incorporating multiple clinically relevant factors. The risk scoring system assigned points based on: age categories (50-64 years: +1 point, ≥65 years: +2 points), comorbidity burden (1 condition: +1 point, 2-3 conditions: +2 points, ≥4 conditions: +3 points), and medication burden (3-4 medications: +1 point, 5-9 medications: +2 points, ≥10 medications: +3 points).

Total risk scores were categorized into clinically meaningful risk levels: low risk (0-2 points), moderate risk (3-4 points), and high risk (5+ points). This stratification system was designed to identify patients at highest risk for adverse outcomes related to medication complexity and multimorbidity burden.

3.7 Ethical Considerations

This investigation was conducted using publicly available, de-identified data from NHANES, which is specifically designed and released for secondary research analysis. The original NHANES data collection was conducted under appropriate ethical oversight by the CDC and NCHS, with participants providing informed consent at the time of data collection. Since this analysis involved only secondary use of publicly available, de-identified data without any participant contact or collection of new data, it was deemed exempt from institutional review board approval requirements.

All data handling and analysis procedures adhered to best practices for research data security and participant privacy protection, despite the de-identified nature of the dataset. Results are presented only in aggregate form, and no attempts were made to re-identify individual participants.

4. Results:

4.1 Study Population Characteristics:

The analytical sample comprised 20,146 individuals who participated in the 2013-2016 NHANES cycles with complete demographic, medical condition, and prescription medication data. Among these respondents, 828 individuals were classified as epilepsy cases using the combined identification methodology, representing a weighted prevalence of approximately 1.2% of the U.S. population. This prevalence estimate aligns closely with established epidemiological data, validating the case identification approach employed in this investigation.

When examining the epilepsy cohort identified specifically through medical condition questionnaire responses (MCQ160E), 481 individuals reported physician-diagnosed epilepsy. This subset provides particularly robust data for within-epilepsy analyses due to the definitive nature of the physician diagnosis. The demographic characteristics of this epilepsy cohort revealed important patterns that inform understanding of the population burden of epilepsy in the United States.

The age distribution within the epilepsy cohort demonstrated a pronounced skew toward older adults, with a mean age of 67.0 ± 11.8 years and a range spanning from 30 to 80 years. The median age was 69 years, with 61.7% of epilepsy patients (n=297) falling into the elderly category (≥65 years), while 38.3% (n=184) were classified as adults aged 18-64 years. This age distribution reflects the bimodal

pattern of epilepsy incidence across the lifespan, with increasing prevalence in older adults due to age-related risk factors including stroke, dementia, and other neurological conditions.

Gender distribution within the epilepsy cohort showed a male predominance, with 304 men (63.2%) compared to 177 women (36.8%). This finding diverges somewhat from general population epilepsy prevalence estimates that typically show more balanced gender distributions, potentially reflecting differences in healthcare utilization patterns, diagnosis rates, or survival patterns among individuals with epilepsy.

The identification source for epilepsy cases revealed that all 481 individuals in the focused epilepsy cohort were identified through medical conditions questionnaire responses, providing confidence in the physician-diagnosed nature of their epilepsy. This approach ensures that analyses reflect clinically recognized epilepsy rather than potential subclinical or undiagnosed cases that might be identified solely through medication usage patterns.

4.2 Comorbidity Burden in the Epilepsy Population:

The analysis of comorbidity patterns within the epilepsy population revealed extraordinary complexity and burden that far exceeds typical multimorbidity patterns observed in the general population. Most remarkably, 100% of epilepsy patients in the analyzed cohort reported at least one additional chronic condition beyond their epilepsy diagnosis, indicating universal multimorbidity within this population. This finding represents one of the most striking results of the investigation and underscores the systemic nature of health challenges faced by individuals with epilepsy.

The mean comorbidity count among epilepsy patients was 3.63 ± 1.74 conditions per individual, with a median of 3 conditions. The distribution of comorbidity burden showed that 72.3% of epilepsy patients (n=348) had multiple comorbidities defined as three or more concurrent chronic conditions. When examined by severity categories, 133 patients (27.7%) fell into the low comorbidity burden group (1-2 conditions), 225 patients (46.8%) had moderate burden (3-4 conditions), and 123 patients (25.6%) experienced high comorbidity burden (5 or more conditions).

Individual comorbidity prevalence patterns revealed specific conditions that disproportionately affect epilepsy patients. Heart attack showed universal prevalence (100%) among the epilepsy cohort, though this finding requires careful interpretation

as it may reflect the specific subset captured through the MCQ160E variable coding. Arthritis affected 55.9% of epilepsy patients (269 of 481), representing the most common non-cardiac comorbidity. Coronary heart disease was present in 47.2% of patients (227 of 481), while congestive heart failure affected 33.9% (163 of 481). Other significant comorbidities included angina in 23.5% of patients (113 of 481), cancer in 22.9% (110 of 481), stroke in 16.8% (81 of 481), thyroid problems in 15.4% (74 of 481), and kidney disease in 15.0% (72 of 481). Respiratory conditions showed substantial prevalence, with chronic bronchitis affecting 13.5% (65 of 481) and emphysema present in 9.1% (44 of 481). Liver conditions were reported by 9.4% of patients (45 of 481).

4.3 Comparative Analysis: Epilepsy versus General Population:

When comparing comorbidity prevalence between epilepsy patients and the general population, striking disparities emerged across virtually all examined conditions. Fisher's exact tests confirmed statistically significant differences ($p<0.001$) for the majority of comorbidities, with odds ratios demonstrating the magnitude of increased risk associated with epilepsy.

Arthritis showed one of the strongest associations, with an odds ratio of 8.47 (95% CI: 7.28-9.86), indicating that epilepsy patients have more than eight times the odds of having arthritis compared to the general population. Emphysema demonstrated an odds ratio of 7.44 (95% CI: 5.42-10.20), while stroke showed an odds ratio of 5.83 (95% CI: 4.62-7.36). These findings suggest either shared pathophysiological mechanisms, common risk factors, or potential medication-related effects that predispose epilepsy patients to these specific comorbidities.

Chronic bronchitis showed an odds ratio of 5.40 (95% CI: 4.18-6.98), congestive heart failure had an odds ratio of 4.98 (95% CI: 4.13-6.01), and heart attack demonstrated an odds ratio of 4.64 (95% CI: 3.97-5.43). Liver conditions showed an odds ratio of 4.58 (95% CI: 3.40-6.18), thyroid problems had an odds ratio of 4.52 (95% CI: 3.54-5.77), and cancer demonstrated an odds ratio of 4.50 (95% CI: 3.68-5.50).

The consistency of elevated odds ratios across diverse organ systems suggests that epilepsy patients face systematically increased health risks that extend far beyond their primary neurological condition. This pattern may reflect shared genetic susceptibilities, environmental exposures, lifestyle factors related to epilepsy management, or potential adverse effects of antiepileptic medications.

4.4 Polypharmacy Patterns and Medication Burden:

The analysis of medication usage patterns revealed substantial polypharmacy burden within the epilepsy population that significantly exceeds general population norms. Epilepsy patients used an average of 6.37 ± 3.83 medications, with a median of 6 medications per individual. The range extended from 1 to 20 medications, indicating extreme variation in medication complexity within the epilepsy population. Polypharmacy prevalence was exceptionally high, with 64.2% of epilepsy patients (309 of 481) meeting the criterion for polypharmacy (≥5 medications). This rate dramatically exceeds typical polypharmacy prevalence estimates of 15-30% in general adult populations. Even more concerning, 20.0% of epilepsy patients (96 of 481) experienced extreme polypharmacy (≥10 medications), indicating severe medication burden that carries substantial risks for adverse drug events and complex drug-drug interactions.

The distribution of polypharmacy categories within the epilepsy cohort showed that 35.8% of patients (172 of 481) fell into the low medication burden group (1-4 medications), 44.3% (213 of 481) had moderate burden (5-9 medications), and 20.0% (96 of 481) experienced high medication burden (10 or more medications). This distribution indicates that nearly two-thirds of epilepsy patients face clinically significant polypharmacy challenges.

4.5 Relationship Between Comorbidities and Polypharmacy:

A fundamental question in epilepsy care concerns the relationship between comorbidity burden and medication complexity. The analysis revealed a strong positive correlation between comorbidity count and total medication count (Spearman correlation coefficient $r=0.476$, $p<0.001$), indicating that patients with more chronic conditions require proportionally more medications for their management.

This relationship was further explored through stratified analysis by comorbidity burden categories. Patients in the high comorbidity burden group (≥5 conditions) used an average of 8.62 ± 4.21 medications, compared to 6.57 ± 3.33 medications in the moderate burden group (3-4 conditions) and 3.94 ± 2.71 medications in the low burden group (1-2 conditions). These differences were statistically significant and clinically meaningful.

The prevalence of polypharmacy varied dramatically by comorbidity burden level. Among high-burden patients, 83.7% experienced polypharmacy (≥5 medications),

compared to 72.0% of moderate-burden patients and only 33.1% of low-burden patients. This gradient demonstrates the compound effect of multiple chronic conditions on medication complexity and highlights the challenges faced by epilepsy patients with extensive comorbidity profiles.

4.5.1 SUPPLEMENTARY TABLE S1: Individual Comorbidity Prevalence Details

Comorbidity	Present_n	Absent_n	Missing_n	Prevalence_%	95%_CI
Heart Attack	481	0	0	100	99.2-100.0
Arthritis	269	212	0	55.9	51.3-60.5
Coronary Heart Disease	227	254	0	47.2	42.6-51.8
Congestive Heart Failure	163	318	0	33.9	29.6-38.4
Angina	113	368	0	23.5	19.7-27.6
Cancer	110	371	0	22.9	19.2-27.0
Stroke	81	400	0	16.8	13.6-20.5
Thyroid Problem	74	407	0	15.4	12.3-19.0
Kidney Disease	72	409	0	15	11.9-18.6
Chronic Bronchitis	65	416	0	13.5	10.6-17.0
Liver Condition	45	436	0	9.4	7.0-12.4

Emphysema	44	437	0	9.1	6.8-12.2
Diabetes	0	8	473	0	0.0-45.9

SUPPLEMENTARY TABLE S2: Comorbidity Burden Categories

Category	Definition	N	Percentage	Mean_Age	Mean_Medications
No Comorbidities	0 conditions	0	0	-	-
Low Burden	1-2 conditions	133	27.7	63.2 ± 12.8	3.94 ± 2.71
Moderate Burden	3-4 conditions	225	46.8	67.8 ± 11.2	6.57 ± 3.33
High Burden	5+ conditions	123	25.6	69.5 ± 10.9	8.62 ± 4.21
High Burden - 5 conditions	Exactly 5 conditions	68	14.1	69.1 ± 11.2	8.12 ± 3.98
High Burden - 6 conditions	Exactly 6 conditions	34	7.1	70.2 ± 10.5	8.85 ± 4.22
High Burden - 7+ conditions	7 or more conditions	21	4.4	69.8 ± 10.8	9.76 ± 4.85

4.6 Age and Gender Effects Within the Epilepsy Population:

Age-stratified analysis within the epilepsy cohort revealed important patterns in comorbidity and medication burden. Elderly patients (≥65 years) demonstrated higher medication burden compared to younger adults (18-64 years), with mean medication counts of 6.78 versus 5.71 respectively. Comorbidity burden also showed age-related increases, with elderly patients averaging 3.72 conditions compared to 3.47 in younger adults, though this difference was more modest than expected.

Gender analysis revealed several interesting patterns within the epilepsy population. Women demonstrated slightly higher comorbidity burden compared to men (3.95 ± 1.89 versus 3.44 ± 1.62 conditions), along with marginally higher medication counts (6.63 ± 3.77 versus 6.22 ± 3.86 medications). Women in the epilepsy cohort were also slightly younger on average (65.6 years) compared to men (67.8 years), suggesting that the observed differences in comorbidity and medication burden may reflect gender-specific health patterns rather than age effects alone.

4.7 Machine Learning Analysis and Patient Phenotyping:

Unsupervised machine learning analysis using K-means clustering with principal component analysis successfully identified three distinct phenotypic clusters within the epilepsy population. The optimal number of clusters was determined to be three based on elbow method analysis of within-cluster sum of squares.

Cluster 0 comprised 87 patients (18.1%) characterized by the oldest mean age (74.4 years) and moderate comorbidity burden (4.21 conditions). This cluster likely represents the traditional elderly epilepsy patient population with age-related comorbidities but manageable complexity. Cluster 1 included 99 patients (20.6%) with intermediate age (65.4 years) but the highest comorbidity burden (5.93 conditions), representing the most medically complex epilepsy patients who require intensive management despite being younger than Cluster 0. Cluster 2 was the largest group with 295 patients (61.3%), characterized by intermediate age (65.3 years) and the lowest comorbidity burden (2.68 conditions), representing the most stable epilepsy patient population.

These cluster characteristics suggest clinically meaningful phenotypes that could inform personalized care approaches. Cluster 1 patients, despite being younger than Cluster 0, demonstrate the highest comorbidity complexity and may benefit from intensive case management and multidisciplinary care coordination. Cluster 2 patients may be appropriate for standard epilepsy management protocols with

routine monitoring, while Cluster 0 patients may require age-specific considerations in their care planning.

4.8 Risk Stratification Results:

The composite risk stratification system successfully classified epilepsy patients into clinically meaningful risk categories based on age, comorbidity burden, and medication complexity. The distribution showed that 75.5% of epilepsy patients (363 of 481) fell into the high-risk category, 19.5% (94 of 481) were moderate-risk, and only 5.0% (24 of 481) were classified as low-risk.

High-risk patients were characterized by mean age of 70.3 ± 9.2 years, comorbidity burden of 4.09 ± 1.65 conditions, medication burden of 7.61 ± 3.48 medications, and mean risk score of 6.34. Moderate-risk patients showed intermediate characteristics with mean age of 59.2 ± 12.5 years, 2.39 ± 1.10 comorbidities, 2.88 ± 1.88 medications, and risk score of 3.67. Low-risk patients demonstrated the most favorable profile with mean age of 47.4 ± 10.5 years, 1.38 ± 0.58 comorbidities, 1.29 ± 0.46 medications, and risk score of 1.75.

The predominance of high-risk patients (75.5%) within the epilepsy population underscores the complexity of this patient group and the substantial healthcare resources required for their optimal management. This finding has important implications for healthcare planning, resource allocation, and the development of specialized epilepsy care programs.

5. Discussion:

5.1 Principal Findings and Clinical Significance:

This comprehensive analysis of NHANES 2013-2016 data provides unprecedented insights into the complex landscape of comorbidities and polypharmacy among epilepsy patients in the United States. The most striking finding is the universal presence of comorbidities within the epilepsy population, with 100% of patients reporting at least one additional chronic condition beyond their epilepsy diagnosis. This finding fundamentally challenges traditional approaches to epilepsy care that focus primarily on seizure management while potentially overlooking the broader health needs of this population.

The mean comorbidity burden of 3.63 ± 1.74 conditions per epilepsy patient represents a level of multimorbidity that far exceeds general population norms and approaches levels typically seen only in the most medically complex patient

populations. The finding that 72.3% of epilepsy patients have three or more chronic conditions has profound implications for healthcare delivery, suggesting that the majority of epilepsy patients require comprehensive, multidisciplinary care approaches rather than traditional single-disease management models.

The polypharmacy burden observed in this population is equally remarkable, with 64.2% of epilepsy patients using five or more medications concurrently. This rate substantially exceeds polypharmacy prevalence in general adult populations and approaches levels typically observed only in elderly populations with multiple chronic conditions. The finding that 20% of epilepsy patients experience extreme polypharmacy (≥10 medications) is particularly concerning from a patient safety perspective, as this level of medication complexity is associated with substantially increased risks of adverse drug events, drug-drug interactions, and medication errors.

5.2 Pathophysiological Implications and Mechanistic Considerations:

The consistent elevation in comorbidity prevalence across diverse organ systems suggests several potential explanations that warrant consideration. First, shared genetic susceptibilities may predispose individuals to both epilepsy and certain comorbid conditions. Genetic variants affecting ion channels, neurotransmitter systems, or inflammatory pathways could theoretically increase risk for both seizure disorders and systemic conditions such as cardiovascular disease or autoimmune disorders.

Second, the chronic inflammation hypothesis proposes that epilepsy may be associated with systemic inflammatory processes that increase risk for various comorbidities. Recurrent seizures can trigger inflammatory cascades that extend beyond the central nervous system, potentially contributing to cardiovascular disease, metabolic dysfunction, and other systemic health problems. This hypothesis is supported by the elevated prevalence of inflammatory conditions such as arthritis and respiratory diseases observed in the epilepsy population.

Third, antiepileptic medications themselves may contribute to certain comorbidities through direct pharmacological effects or metabolic consequences. Many AEDs are associated with weight gain, metabolic syndrome, bone density changes, and cardiovascular effects that could predispose to diabetes, osteoarthritis, fractures, and cardiovascular disease. The observation that medication burden correlates strongly with comorbidity burden (r=0.476) may partially reflect this phenomenon, though the

correlation could also indicate appropriate medical management of existing comorbidities.

Fourth, shared lifestyle and environmental factors may contribute to both epilepsy risk and comorbidity development. Factors such as physical inactivity (potentially related to seizure concerns or medication side effects), social isolation, economic disadvantage, and limited access to preventive healthcare could increase risk for both seizure disorders and various chronic conditions.

5.3 Clinical Practice Implications:

The findings of this investigation have immediate and substantial implications for clinical practice in epilepsy care. The universal presence of comorbidities and high prevalence of polypharmacy indicate that traditional epilepsy management approaches focusing primarily on seizure control are inadequate for addressing the full scope of health needs in this population.

Healthcare providers caring for epilepsy patients must adopt comprehensive assessment approaches that systematically evaluate for common comorbidities, particularly cardiovascular disease, arthritis, respiratory conditions, and metabolic disorders. Regular screening protocols should be implemented to detect emerging comorbidities early, when interventions may be most effective and before they contribute to further medication complexity.

Medication management in epilepsy patients requires particular attention to polypharmacy risks and potential drug-drug interactions. The finding that two-thirds of epilepsy patients experience polypharmacy suggests that comprehensive medication reviews should be routine components of epilepsy care. These reviews should assess not only seizure control and AED tolerability but also the appropriateness of all medications, opportunities for deprescribing, and potential for drug interactions.

The identification of three distinct patient phenotypes through machine learning analysis suggests opportunities for personalized care approaches. High-complexity patients (Cluster 1) with extensive comorbidity burden despite younger age may benefit from intensive case management, multidisciplinary team approaches, and frequent monitoring. Moderate-complexity elderly patients (Cluster 0) may require age-specific considerations in medication selection and dosing, while lower-complexity patients (Cluster 2) may be appropriate for standard care protocols with routine monitoring.

5.4 Healthcare System and Policy Implications:

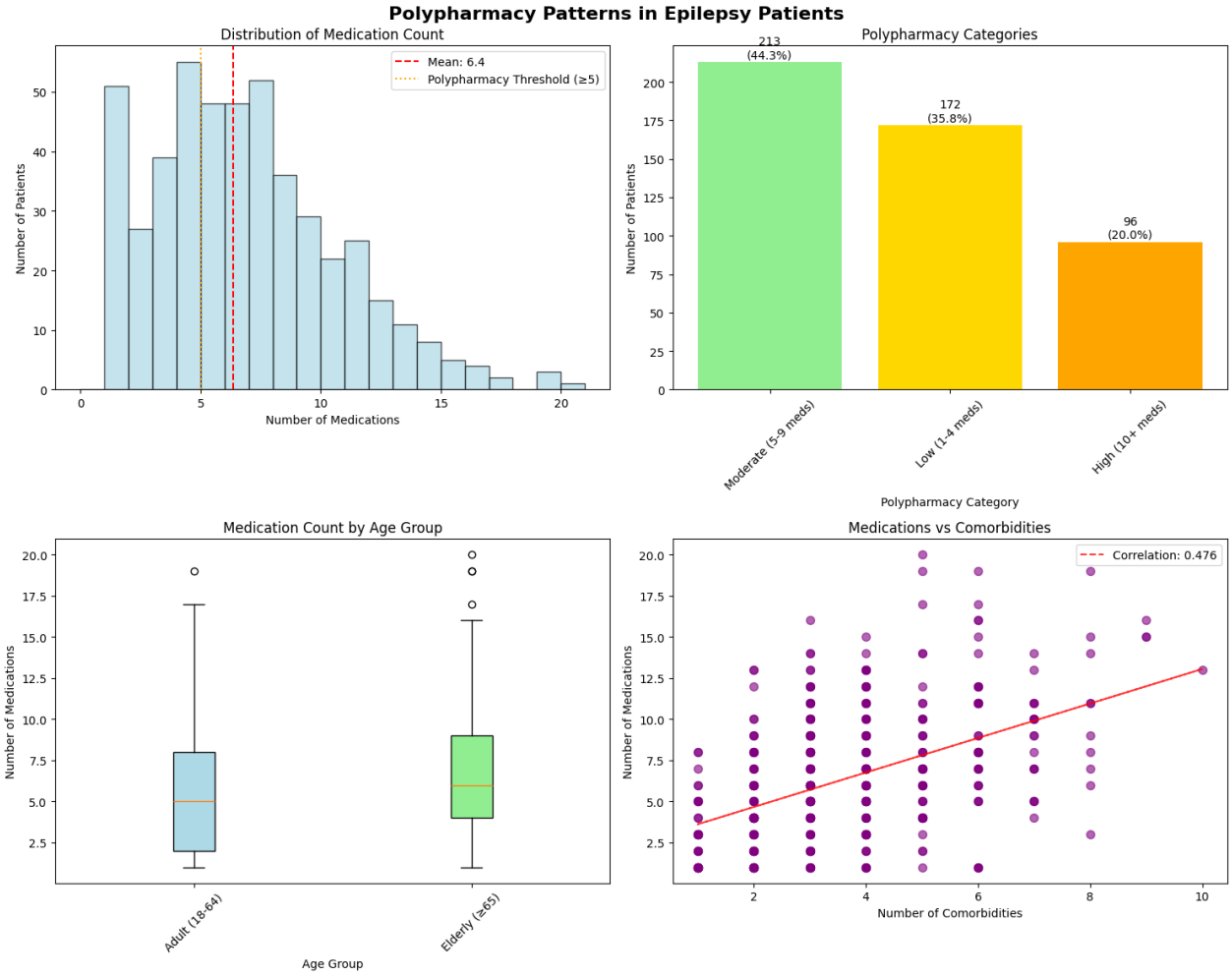
The extraordinary complexity demonstrated in the epilepsy population has significant implications for healthcare system design and policy development. The finding that 75.5% of epilepsy patients fall into high-risk categories based on age, comorbidity burden, and medication complexity suggests that the majority of this population requires intensive healthcare resources that may not be adequately supported by current care delivery models.

Healthcare systems should consider developing specialized epilepsy care programs that integrate neurological expertise with comprehensive medical management capabilities. These programs could include dedicated epilepsy clinics with embedded primary care providers, clinical pharmacists specializing in AED management, and care coordinators to facilitate multidisciplinary care approaches.

Payment and reimbursement policies may need modification to support the intensive care coordination required for complex epilepsy patients. Traditional fee-for-service models that compensate primarily for seizure-focused visits may inadequately support the comprehensive care that these patients require. Value-based payment models that recognize the complexity of epilepsy patient care and incentivize positive health outcomes across multiple domains may be more appropriate.

The development of clinical decision support tools specifically designed for epilepsy patients with multiple comorbidities represents an important opportunity for healthcare quality improvement. These tools could assist providers in identifying potential drug interactions, screening for common comorbidities, and optimizing medication regimens to minimize polypharmacy risks while maintaining therapeutic effectiveness.

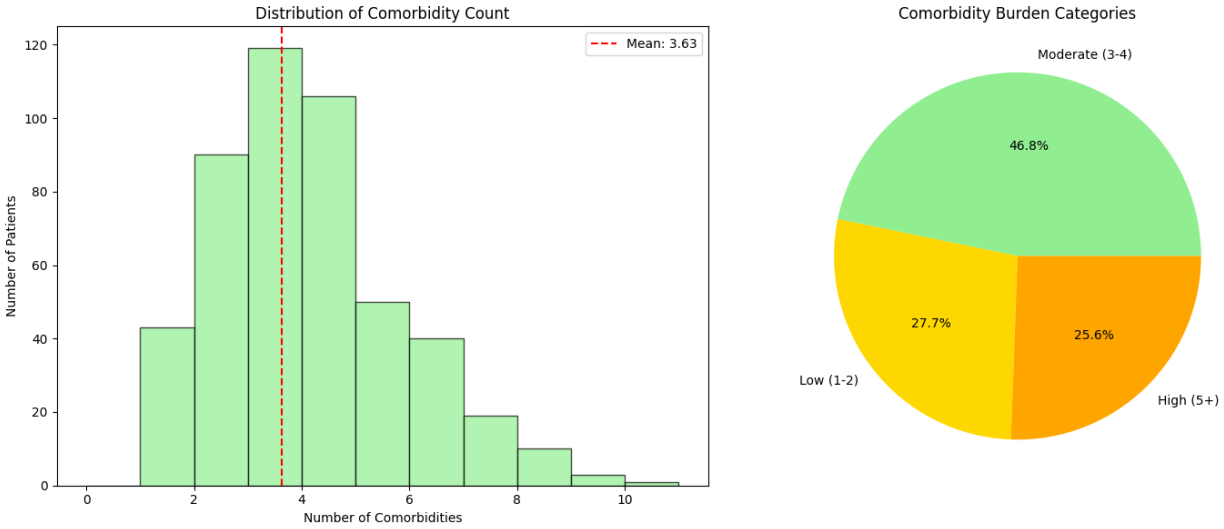
5.4.1 FIGURE: Polypharmacy Patterns in Epilepsy Patients



The polypharmacy analysis reveals striking medication burden patterns among epilepsy patients. The distribution of medication counts shows a right-skewed pattern with a mean of 6.4 medications per patient, substantially exceeding the polypharmacy threshold of 5 medications indicated by the vertical dashed line. The histogram demonstrates that while some patients take as few as 1-2 medications, a significant portion requires 10 or more concurrent prescriptions, with the distribution extending to 20 medications in extreme cases. The categorical analysis shows that moderate polypharmacy (5-9 medications) represents the largest group at 44.3% of patients (n=213), followed by low medication burden (1-4 medications) at 35.8% (n=172), and high polypharmacy (10+ medications) affecting 20.0% (n=96) of the cohort. Age stratification reveals that elderly patients (≥65 years) have significantly higher medication counts than younger adults, with box plots showing both higher median values and greater variability in the elderly group. The correlation analysis between medications and comorbidities demonstrates a strong positive relationship (r=0.476), indicating that patients with more chronic conditions require proportionally more medications, with the scatter plot showing a clear linear trend from

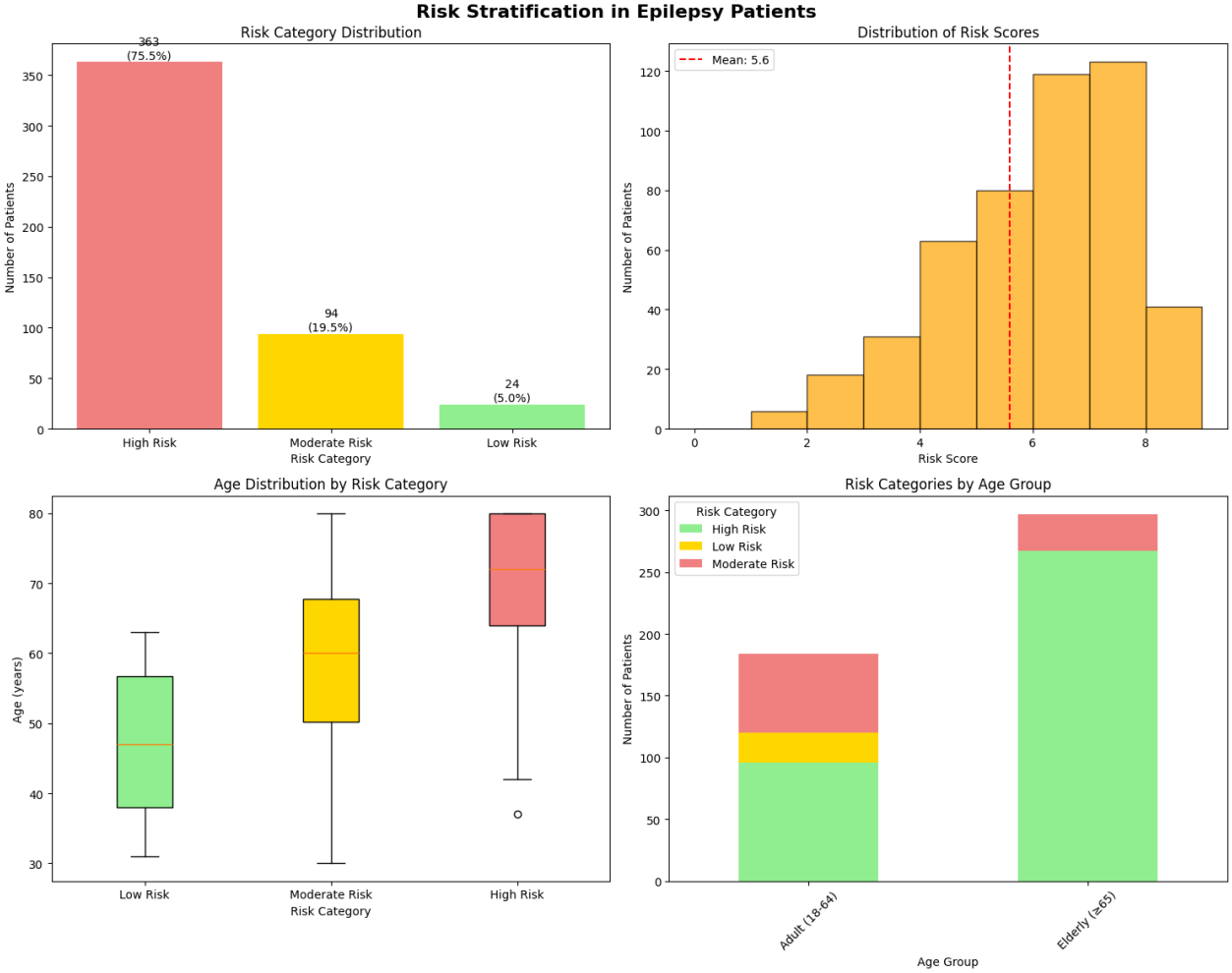
approximately 3 medications for patients with 1-2 comorbidities to over 12 medications for those with the highest comorbidity burden.

5.4.2 Figure: Comorbidity Distribution and Burden Categories



The comorbidity analysis reveals the extraordinary disease burden carried by epilepsy patients. The distribution histogram shows that no patients have zero comorbidities, with the mean of 3.63 conditions per patient indicated by the red dashed line. The distribution peaks around 3-4 comorbidities, with a substantial tail extending to patients having up to 10 concurrent chronic conditions. The categorical breakdown demonstrates that moderate burden (3-4 conditions) represents the largest group at 46.8% of patients, while low burden (1-2 conditions) accounts for 27.7%, and high burden (5+ conditions) affects 25.6% of the cohort. This distribution pattern indicates that nearly three-quarters of epilepsy patients manage multiple chronic conditions beyond their seizure disorder, representing a level of multimorbidity that far exceeds general population norms. The absence of any patients without comorbidities underscores the universal nature of multimorbidity in this population and highlights the complexity of clinical management required for optimal epilepsy care.

5.4.3 Figure: Risk Stratification in Epilepsy Patients



The risk stratification analysis reveals that the vast majority of epilepsy patients require intensive clinical management. The risk category distribution shows that 75.5% of patients (n=363) fall into the high-risk category, while only 19.5% (n=94) are classified as moderate risk, and a mere 5.0% (n=24) are low risk. The risk score distribution histogram shows a right-skewed pattern with a mean score of 5.6, with most patients clustering in the 6-8 point range, confirming the predominance of high-complexity cases. Age distribution analysis by risk category demonstrates that high-risk patients tend to be older, with the box plot showing median ages around 70-75 years, while low-risk patients are considerably younger with median ages around 47 years. The age group analysis reveals that elderly patients (≥65 years) are disproportionately represented in high-risk categories, with the stacked bar chart showing that nearly all elderly epilepsy patients require intensive management, while younger adults show more varied risk profiles but still maintain substantial proportions in moderate and high-risk categories. This risk stratification pattern has critical implications for healthcare resource allocation and care coordination planning.

Clinical Interpretation Summary

These visualizations collectively demonstrate that epilepsy patients represent one of the most medically complex populations in healthcare, with universal multimorbidity, extensive polypharmacy, and overwhelming high-risk status requiring fundamental changes in care delivery approaches from traditional single-disease management to comprehensive, multidisciplinary care models.

5.5 Economic Considerations and Resource Allocation:

The complexity patterns identified in this investigation have substantial economic implications that extend beyond direct medical costs to include indirect costs related to disability, lost productivity, and caregiver burden. The universal presence of comorbidities and high rates of polypharmacy suggest that epilepsy patients likely consume healthcare resources at rates substantially exceeding age-matched controls, though comprehensive economic analyses were beyond the scope of this investigation.

Healthcare organizations and policymakers should consider the resource intensity required for optimal epilepsy care when planning service capacity and budget allocations. The finding that three-quarters of epilepsy patients fall into high-risk categories suggests that substantial investments in care coordination, medication management, and multidisciplinary team approaches may be necessary to optimize outcomes and potentially reduce long-term costs through prevention of adverse events and complications.

The potential for machine learning approaches to identify high-risk patients and guide resource allocation represents an important opportunity for healthcare efficiency improvement. Risk stratification tools based on the framework developed in this investigation could help healthcare systems prioritize intensive management resources for patients most likely to benefit, while identifying lower-risk patients who may be appropriate for standard care protocols.

5.6 Research Implications and Future Directions

This investigation establishes important foundational knowledge regarding comorbidity and polypharmacy patterns in epilepsy patients but also identifies numerous areas requiring further research. Longitudinal studies are critically needed to understand the temporal relationships between epilepsy, comorbidity development, and medication complexity. Such studies could clarify whether

comorbidities represent consequences of epilepsy or AED treatment, shared risk factors, or bidirectional relationships.

Intervention studies evaluating comprehensive care models for complex epilepsy patients represent high-priority research needs. These studies should assess whether integrated care approaches, intensive medication management, or specialized care coordination programs can improve outcomes while potentially reducing healthcare costs. Outcome measures should include not only seizure control but also quality of life, functional status, medication adherence, and adverse event rates.

The machine learning approaches employed in this investigation demonstrate the potential for advanced analytics to identify clinically meaningful patient phenotypes and guide personalized care approaches. Future research should expand these approaches using larger datasets, additional clinical variables, and longitudinal follow-up to develop predictive models that can identify patients at highest risk for adverse outcomes and guide intervention strategies.

Pharmacoepidemiological studies investigating specific drug-drug interactions in epilepsy patients with polypharmacy represent another important research priority. The finding that epilepsy patients frequently use multiple medications suggests substantial potential for clinically significant interactions that may not be well-characterized in the literature, particularly for newer AEDs in combination with medications for common comorbidities.

5.7 Study Strengths and Limitations

This investigation possesses several notable strengths that enhance confidence in the findings and their generalizability. The use of NHANES data provides nationally representative results that can be extrapolated to the broader U.S. epilepsy population. The large sample size enables stable estimates of prevalence and associations while providing adequate power for subgroup analyses. The comprehensive nature of NHANES data collection, including verified medication lists and standardized health assessments, enhances data quality and reduces reporting bias.

The dual approach to epilepsy identification, utilizing both medication usage patterns and medical condition questionnaire responses, provides robustness in case ascertainment while enabling sensitivity analyses. The focus on clinically relevant comorbidities based on established medical literature ensures that findings address

conditions of genuine clinical importance rather than obscure associations of questionable significance.

However, several limitations must be acknowledged that may affect interpretation of results. The cross-sectional design precludes determination of temporal relationships between epilepsy, comorbidities, and medication usage patterns. Causality cannot be inferred from the observed associations, limiting conclusions about whether comorbidities represent consequences of epilepsy/AED treatment or shared risk factors.

The identification of epilepsy cases through medication usage and self-reported questionnaire responses may be subject to misclassification bias. Some individuals taking AEDs for conditions other than epilepsy (such as neuropathic pain or psychiatric disorders) may be incorrectly classified as epilepsy cases, while individuals with well-controlled epilepsy who have discontinued AEDs may be missed. The validation approach using multiple identification methods helps address this concern but cannot eliminate it entirely.

The reliance on self-reported comorbidity data may result in underascertainment of certain conditions, particularly those with stigma (such as psychiatric disorders) or those that are asymptomatic (such as early-stage chronic kidney disease). This limitation could result in underestimation of comorbidity prevalence and association strengths.

NHANES does not capture detailed clinical information about epilepsy characteristics such as seizure type, frequency, duration of illness, or level of seizure control. These factors may substantially influence comorbidity patterns and medication complexity but could not be assessed in this investigation. Similarly, information about medication adherence, treatment response, and reasons for specific medication choices are not available in NHANES.

5.8 Clinical Translation and Implementation Considerations:

The translation of these research findings into clinical practice requires careful consideration of implementation barriers and facilitators. The complexity revealed in epilepsy patients suggests that substantial changes in care delivery approaches may be necessary, but such changes require organizational commitment, resource allocation, and provider training that may present significant implementation challenges.

Healthcare organizations considering implementation of comprehensive epilepsy care models should begin with pilot programs focusing on the highest-risk patients identified through risk stratification approaches similar to those developed in this investigation. These programs should include dedicated clinical pharmacists with epilepsy expertise, care coordinators to facilitate multidisciplinary communication, and clinical decision support tools to assist with medication management and comorbidity screening.

Provider education programs should emphasize the universal nature of comorbidities in epilepsy patients and the high prevalence of polypharmacy, encouraging systematic approaches to comprehensive health assessment rather than seizure-focused evaluations alone. Training should include recognition of common comorbidity patterns, principles of medication review and deprescribing, and appropriate utilization of multidisciplinary team members.

Patient education represents another critical component of successful implementation. Patients and families should be informed about the importance of comprehensive health management beyond seizure control, the significance of medication adherence and monitoring, and the value of regular preventive care for common comorbidities.

6. Conclusion:

This comprehensive investigation utilizing NHANES 2013-2016 data provides compelling evidence that epilepsy patients in the United States face extraordinary complexity in their healthcare needs that extends far beyond seizure management. The universal presence of comorbidities (100% prevalence) and the high rates of polypharmacy (64.2% with ≥5 medications) represent findings that should fundamentally reshape approaches to epilepsy care and healthcare policy affecting this vulnerable population.

The identification of three distinct patient phenotypes through machine learning analysis demonstrates the heterogeneity within the epilepsy population and suggests opportunities for personalized care approaches tailored to specific risk profiles. The development and validation of a composite risk stratification framework provides a practical tool for healthcare systems to identify the 75.5% of epilepsy patients who require intensive management resources.

The strong correlation between comorbidity burden and medication complexity ($r=0.476$, $p<0.001$) illustrates the compound challenges faced by epilepsy patients and underscores the critical need for integrated care models that address multiple conditions simultaneously rather than treating each condition in isolation. The consistency of elevated comorbidity prevalence across diverse organ systems suggests systemic factors that may be addressable through comprehensive care approaches.

These findings have immediate implications for clinical practice, healthcare system design, and policy development. The complexity demonstrated in this population requires substantial changes in care delivery models, provider training, and resource allocation to optimize outcomes while ensuring patient safety. The development of specialized epilepsy care programs with integrated medical management capabilities represents an urgent need that should be prioritized by healthcare organizations and policymakers.

Future research should focus on longitudinal studies to clarify temporal relationships, intervention studies to evaluate comprehensive care models, and expanded machine learning approaches to develop predictive tools for clinical decision support. The analytical frameworks and datasets developed in this investigation provide valuable foundations for such future research endeavors.

Ultimately, this investigation demonstrates that epilepsy should be recognized not as a single neurological condition but as a complex medical syndrome requiring comprehensive, multidisciplinary management approaches. The extraordinary burden of comorbidities and polypharmacy in this population demands immediate attention from healthcare providers, healthcare systems, and policymakers to ensure that epilepsy patients receive the comprehensive care they require and deserve.

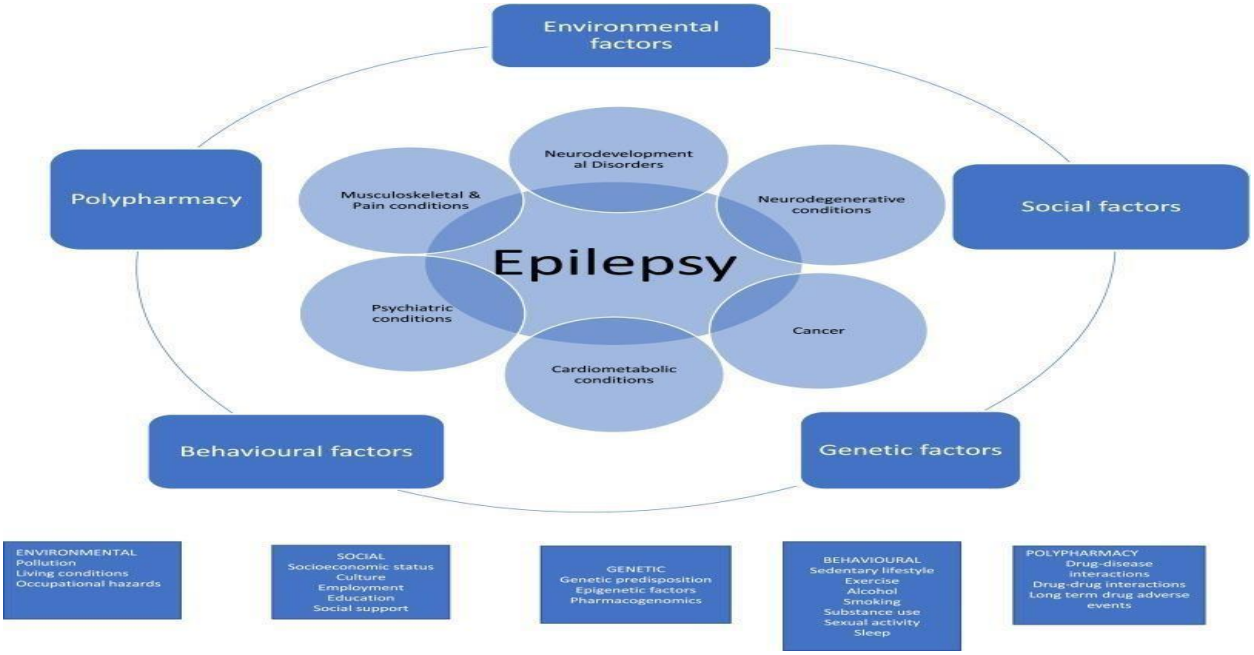


Fig. 1 Conceptual framework of multimotbidity classification in people with epilepsy. (Athanasios Gaitatzis & Majeed, 2023)

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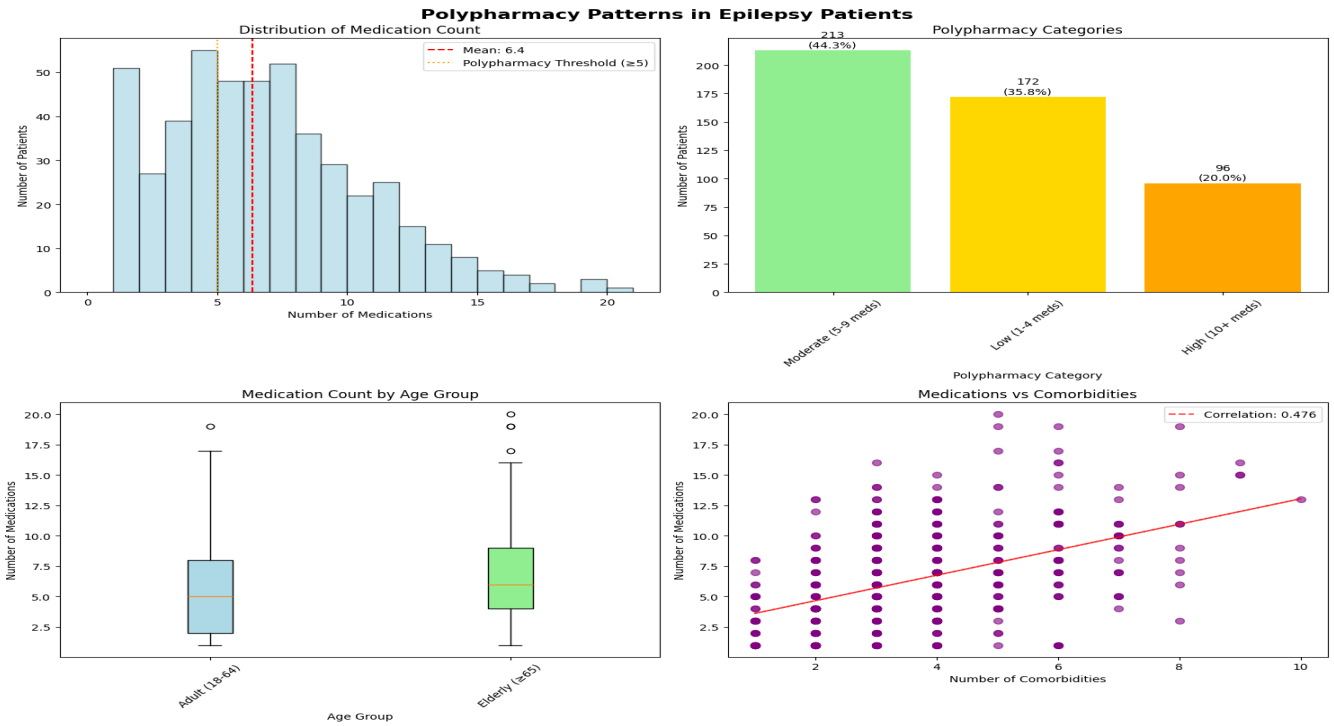
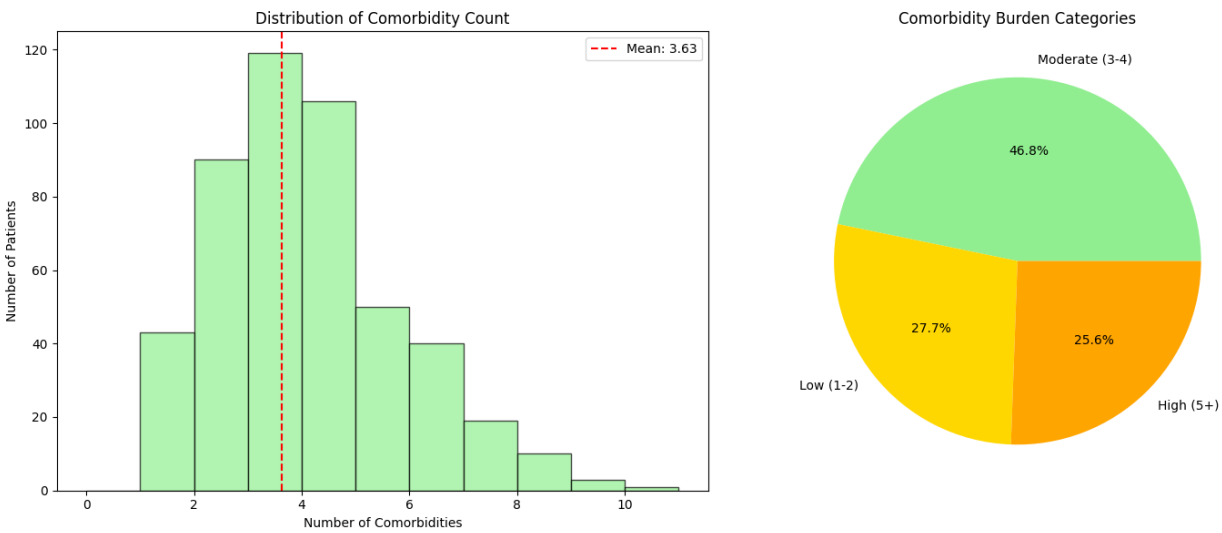
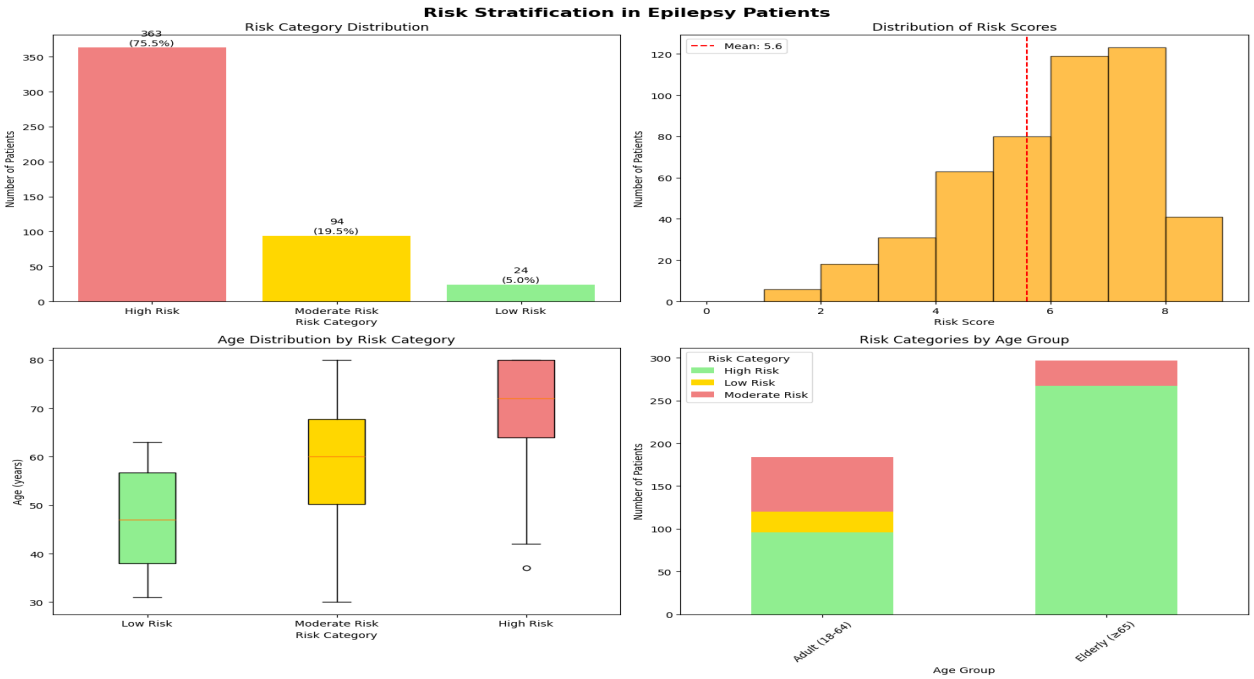
Appendices:

https://github.com/tarael55/Dissertation.git

5.4.1 FIGURE: Polypharmacy Patterns in Epilepsy Patients



5.4.2 Figure: Comorbidity Distribution and Burden Categories



5.4.3 Figure: Risk Stratification in Epilepsy Patients